

Mean Squared Error

- ❑ Mean Squared Error (MSE) – SSE/n where n is the number of instances in the data set
 - ❑ This can be nice because it normalizes the error for data sets of different sizes
 - ❑ MSE is the average squared error per pattern
- ❑ Root Mean Squared Error (RMSE) – is the square root of the MSE
 - ❑ This puts the error value back into the same units as the features and can thus be more intuitive
 - ❑ Since we squared the error on the SSE
 - ❑ RMSE is the average distance (error) of targets from the outputs in the same scale as the features

Perceptron Learning Theorem

- A perceptron (threshold unit) can *learn* anything that it can *represent* (i.e. anything linearly separable with a hyperplane)

OR function

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1

